



Palo Alto Networks Cybersecurity Academy – Configuring GlobalProtect RAVPN on a PA220 Firewall

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Purpose

This lab introduces how to configure a VPN using a Palo Alto 220 firewall with the built in GlobalProtect. This lab builds the skills on how to set up a remote access VPN meaning if a globally routable address is used as the portal to the VPN, internal networks connected to the Palo Alto 220 will be accessible. This VPN uses IPsec, an encryption level that is used in enterprise environments. This lab builds skills that are invaluable to the cooperative world where work is moving to the employee's homes and provides a way to access sensitive cooperative data from anywhere without compromising data integrity.

Background

As mentioned in the purpose section, more companies since the COVID-19 epidemic have shifted to work in home environments. The bureau of labor statistics shows that remote work has increased by 30% across all industries within two years of the covid pandemic. And utilizing sensitive company information without a secure method of encrypting the data poses a risk of that potentially propriety of information leaking and distributed globally. A solution to this problem of unsecure remote work is through a Virtual Private Network or a VPN for short. A VPN provides companies with a secure way to both access and transmit corporate data through a secure encryption.

It is estimated that 1.75 billion people globally have used a VPN before. This is done through a VPN client, a software that enables the user to contact the VPN server to initialize and maintain the encryption. For this lab the software is downloaded once the VPN server's globally routable IP address is reached, typically to connect to a VPN server no IP address is necessary as the VPN client fetches the VPN server IPs and will connect the user to the fastest server or a server in the country the user specifies.

Once the server is selected using the VPN client the user's machine sends a TLS (Transport Layer Security) encrypted packet containing information about what encryption protocols the client can run like TLS versions and cipher suites (Encryption Algorithms). The server then responds to the "ClientHello" with a ServerHello" choosing all the most secure protocols the client can run. Immediately after the "SeverHello" is sent, the server sends a public certificate to verify the authenticity of the VPN server. This certificate is trusted by your computer because it is signed by a trusted CA or Certificate authority, and it can only be signed by a CA using a private key making anything the CA signs trustworthy. Your operating system comes with a list of trusted Certificate Authorities giving it the ability to verify and protect your system from rouge servers trying to establish connections to a client's machine. Once the VPN server is verified to be trustworthy, both the client and the server will use asymmetric encryption.

Asymmetric encryption works by using a public key which anyone can see and both client and server using their private key to create a shared secret. This is possible with the Duffie-Hellman, which is a mathematical algorithm which is easy to do, but not to undo. To simplify, if person A and person B both chose a number (this represents the private key), A chose 9 and B chose 3 and there was also a number everyone agreed on (the public key), everyone chose 8. Person A and B take the agreed number and raise it to the power of their secret number. A now has 8^9 and B has 8^3 ; they then send that number to each other and raise that number again to their own secret number. This will result in both parties having the same number; this would be the shared secret that verifies that they have secure asymmetric communication with each other. The real Duffie-Hellman used in modern TLS handshakes are much more complicated and secure, making the math of undoing the calculation impossible with current tech.

Once both client and VPN server have the same shared secret, they then use that shared secret along with the client's random number and the server's random number they sent to each other in their "ClientHello" and "ServerHello" messages to derive a session key with the same KDF (Key Derivation Function). Once both sides have the session key, they put the key along with all handshake messages ("ClientHello" and "ServerHello") into a "verify_data" variable which is hashed using HMAC with the shared secret key. Once all the data on is packaged in "verify_data" the clients send a "Finished" message that is encrypted by the session key once the server receives this message and matches it with its own "verify_data" variable it can then send its "Finished" message to the client. After both sides verify each other's packets, they can then send encrypted messages to each other across the internet.

Once encrypted, communication is secured by a tunnel established between the client and the VPN server. A tunnel is when a client and server are connected across the internet, and data can only be accessed by client and server. Once the packets are encrypted, the internet can only see the globally routable IP address of the client router and the globally routable IP address of the VPN server. Once this tunnel is established, the client is assigned to a private tunnel IP address. This IP address is considered inside the network of the firewall/VPN server as it can route those addresses to its own network. And once you are inside the network, your global address now looks like the IP address of the VPN server, making your IP address look like wherever the server is from. This is how VPNs disguise your address to be from another country, and it is also how a VPN can extend a private network allowing employees to access resources from within the company network by connecting to a VPN to receive a tunnel IP that is considered within the company network.

This is how a TLS VPN works, but the security protocols used for the GlobalProtect VPN are IPsec and TLS which follow the same fundamental concepts and principles but use different methods of creating keys and encrypting packets.

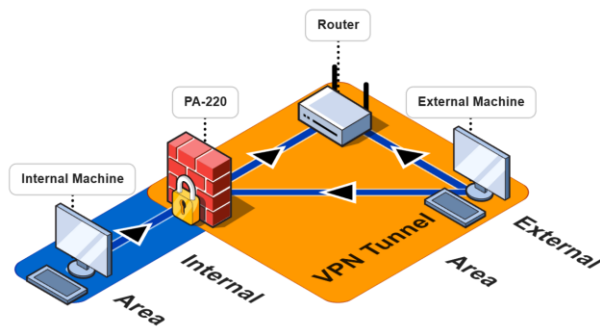
GlobalProtect uses IPsec for the VPN tunnel but uses HTTPS inside TLS to authenticate the client and gives the client a list of gateways to connect to. When using Globalprotect for a VPN, there is a portal, which is used specifically for client authentication and sending the client configurations for gateways. This TLS encryption for connecting to the portal protects user credentials like usernames and passwords. Client certificates, and any other way a client can prove their identity to the portal. A GlobalProtect Gateway can be thought of as the VPN server but uses IPsec instead of TLS.

IPsec works very similarly to TLS but has more keys and more tunnels that are dedicated to transport specific information. Just like TLS, IPsec uses the Diffie-Hellman to get their shared secrets, then combines that with the nonces (random number) of the client and server to create a SKEYSEED from this a protocol called Internet Key Exchange or IKE derives multiple keys, one for key derivation, authentication, encryption, and integrity. From these keys, a secure control channel is created, called IKE SA. This entire process is known as IKE_SA_INIT, in the control channel that is created for any management information to do with the VPN tunnel that will be created for user traffic will be sent and encrypted. This could include ending the session, new configurations, and more. Using the newly created IKE SA, the client GlobalProtect Gateway exchange certificates, verify signatures, and authenticate each other to then create the tunnel for client traffic.

To create the VPN tunnel for client traffic, even more keys are derived. The Key derivation key created from the SKEYSEED alongside more nonces, and more information is used to create KEYMAT which is then used to derive two more keys, one ESP key for encryption and another for authentication. Using these ESP keys, a data tunnel containing all client traffic is encrypted using the encryption key. This tunnel specifically for user data is called the ESP SA. Just like the TLS VPN one the user is connected to the Gateway, they are assigned a tunnel IP address which is an internally routable address allowing remote users to access information within the network from anywhere.

Being able to understand the inner workings of a VPN help significantly improve troubleshooting capabilities and give a greater depth of understanding of VPNs and how encryption works. Having a Remote Access VPN allows companies to effectively sustain remote employees and reduce costs related to office sizes and maintenance while maintaining the level of security employees would get from working in the office.

Network Diagram



Lab Summary

In this lab, I configured a GlobalProtect Remote Access VPN in a Palo Alto 220 Firewall. The VPN was configured with credentials that needed to be known beforehand to login to the VPN portal and download the software client. Once a software client is downloaded, the portal IP is entered and will connect you to the Gateway, and internal resources can be accessed.

Lab Commands

Creating Certificates

The first step in creating a VPN is to create certificates; this ensures secure communication between the client and server as a means of authentication.

Navigate to Device > Certificates > Create

First, a root certificate is needed to sign off on the server certificate, you a root certificate is already created, you can skip this step.

Generate Certificate?

Certificate Type

☒ Local ☐ SCEP

Certificate Name

Root-Cert

Common Name

Root-Cert

IP or FQDN to appear on the certificate

Signed By

☒ Certificate Authority

☐ Block Private Key Export

OCSP Responder

Cryptographic Settings

Algorithm

RSA

Number of Bits

2048

Digest

sha256

Expiration (days)

365

Certificate Attributes

	TYPE	VALUE
☐	Country = "C" from "Subject" field	US
☒	Organization = "O" from "Subject" field	Palo Alto Networks

+ Add

- Delete

Generate

Cancel

Make sure the Certificate Authority box is checked; this is what makes this certificate a Certificate Authority or Root Certificate

Next, create a Server Level Certificate; this is the certificate sent for authentication when the client is connecting to the portal to authenticate it.

Generate Certificate

Certificate Type
☒ Local
☐ SCEP

Certificate Name
Server-Cert

Common Name
192.168.40.167

IP or FQDN to appear on the certificate

Signed By
Root-Cert

☐ Certificate Authority
☐ Block Private Key Export

OCSP Responder

Cryptographic Settings

Algorithm
RSA

Number of Bits
2048

Digest
sha256

Expiration (days)
365

Certificate Attributes

TYPE	VALUE
☒ Host Name = "DNS" from Subject Alternative Name (SAN) field	192.168.40.127
☐ IP = "IP Address" from Subject Alternative Name (SAN) field	192.168.40.127
☐ Organization = "O" from	Palo Alto Networks

Add
Delete

Generate
Cancel

When creating the server level certificate ensure that the common name field is the IP address of the firewalls outside interface. This common name is used to authenticate and verify that the firewall matches the IP address the client is trying to connect to. Your ISP that gives the firewalls its address may assign a different IP address to the firewall occasionally, and in that case the common name will need to be recreated or edited if possible.

Next, the TLS/SSL profile is configured; this tells the firewall that TLS methods it can use to connect with the client. If you want your connection to only accept the most secure methods of TLS, then only the latest versions should be selected for this TLS/SSL service profile configuration.

SSL/TLS Service Profile

Name
VPN_SSL/TLS_Profile

Certificate
Server-Cert

Protocol Settings

Min Version
TLSv1.0

Max Version
Max

OK
Cancel

Make sure when creating the profile to assign the Server Certificate to it; the root certificate is only used to sign off on lower-level certificates.

Next, create a certificate profile; this will be needed later when configuring the portals and gateways.

This is the tab right under the tab used to create the certificates.

Certificate Profile

CA Certificate: Root-Cert

Default OCSP URL:

OCSP Verify Certificate: None

Template Name/OID:

OK Cancel

Certificate Profile

Name: vpn_profile

Username Field: None

User Domain:

CA Certificates

☐	NAME	DEFAULT OCSP URL	OCSP VERIFY CERTIFICATE	TEMPLATE NAME/OID
☐	Root-Cert			

+ Add - Delete ↑ Move Up ↓ Move Down

Default OCSP URL (must start with http:// or https://)

☐ Use CRL CRL Receive Timeout (sec) 5

☐ Use OCSP OCSP Receive Timeout (sec) 5

OCSP takes precedence over CRL Certificate Status Timeout (sec) 5

☐ Block session if certificate status is unknown

☐ Block session if certificate status cannot be retrieved within timeout

☐ Block session if the certificate was not issued to the authenticating device

☐ Block sessions with expired certificates

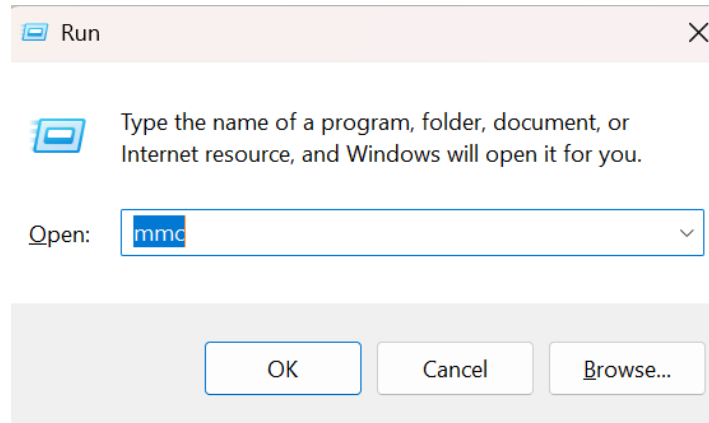
OK Cancel

Next, the certificates need to be installed on the machine so that the computer trusts the device on the network. In this lab, the certificates were installed on the machine.

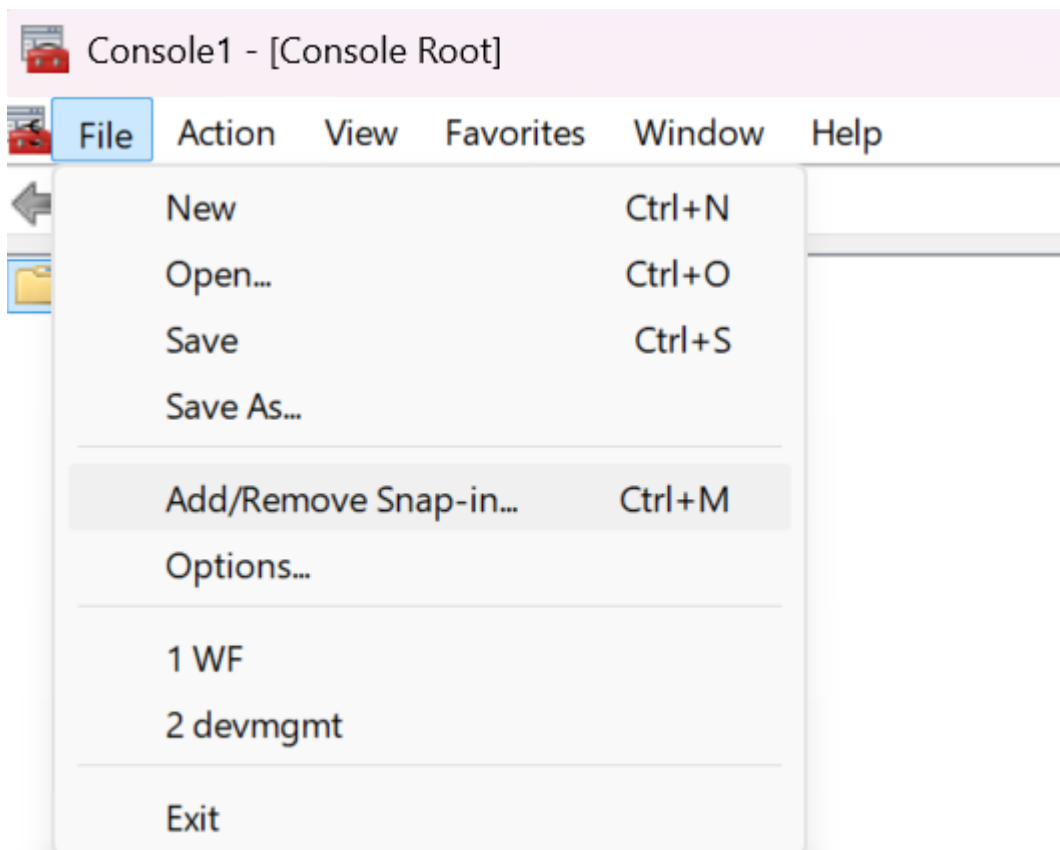
Go back to Device > Certificates and export both certificates, the button for this action is next to the generate button.

Hit Windows + R and type mmc, hit enter.

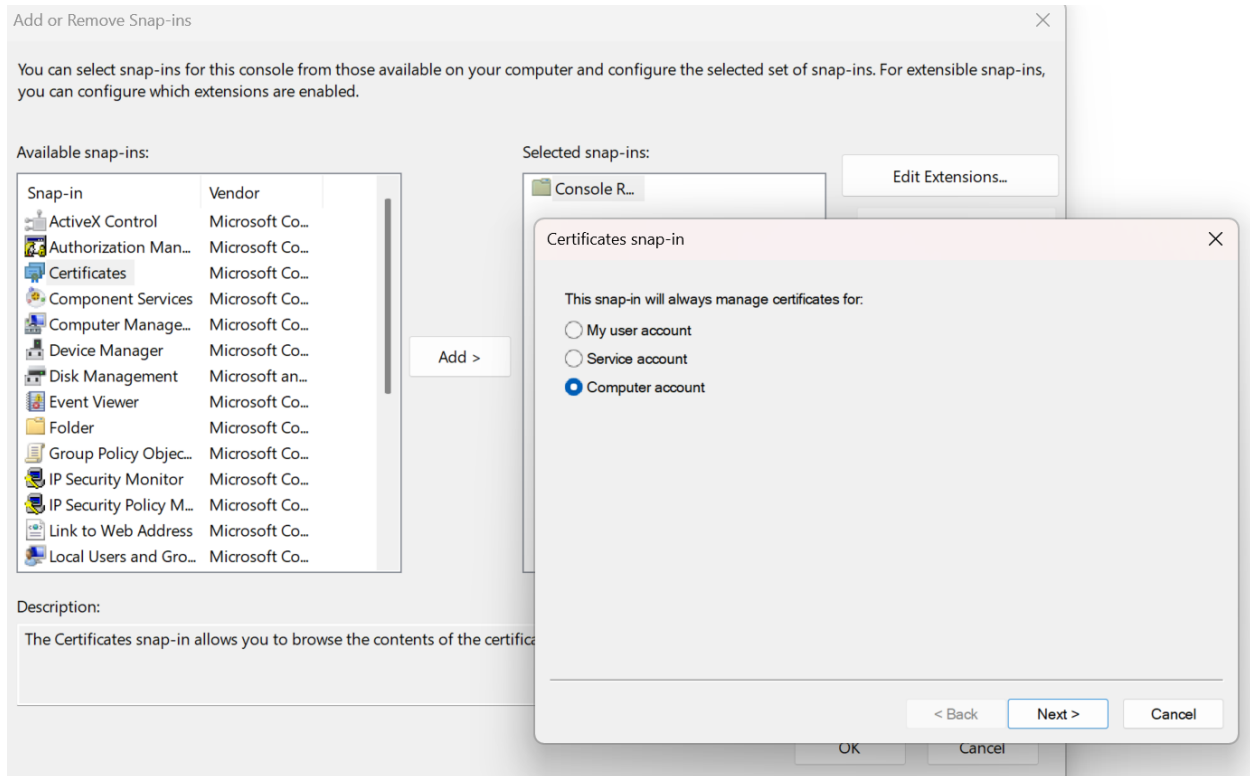
This takes you to the Microsoft Management Console; this will be used to install certificates.



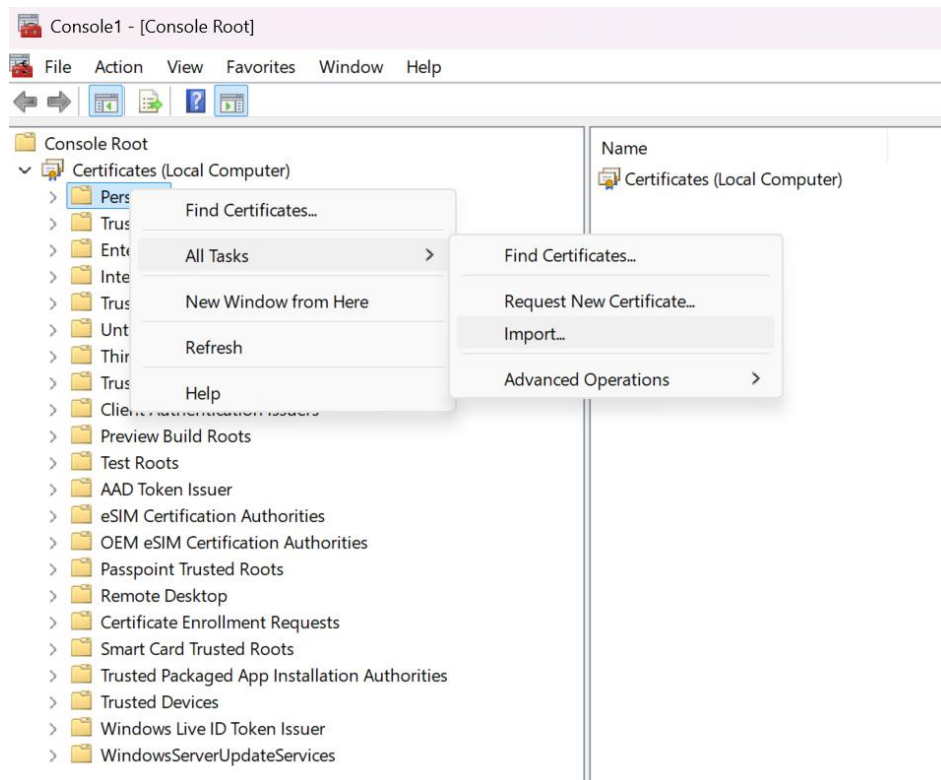
Once in the management console, navigate to files in the top right corner and select “Add/Remove Snap-in”



Next, the management console will prompt you to choose a snap in, choose Certificates, and where you want to install it.



Once you have added the snap in to find the personal folder, right click it, select all tasks, and choose to import.



You will then be taken to the Certificate Import Wizard and be prompted to select a certificate to import, select only the server certificate to import.

You will then be prompted to select a folder to import it to, but it should already have selected the personal folder, once confirmed click next and finish.

←  Certificate Import Wizard

×

Certificate Store

Certificate stores are system areas where certificates are kept.

Windows can automatically select a certificate store, or you can specify a location for the certificate.

☐ Automatically select the certificate store based on the type of certificate

☒ Place all certificates in the following store

Certificate store:

Personal

Browse...

Next

Cancel

←  Certificate Import Wizard

Completing the Certificate Import Wizard

The certificate will be imported after you click Finish.

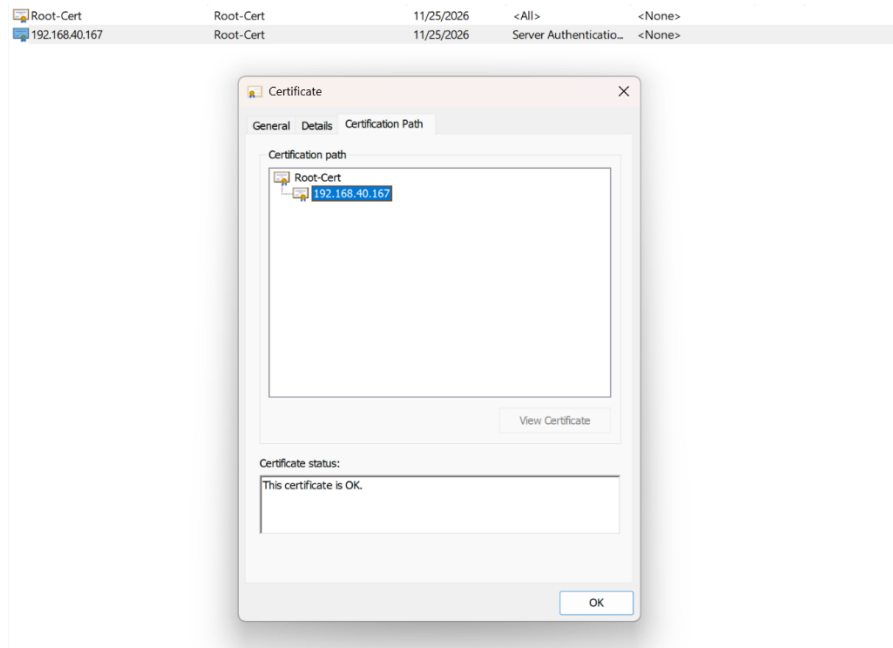
You have specified the following settings:

Certificate Store Selected by User	Personal
Content	Certificate
File Name	C:\Users\kacin\Downloads\cert_Server-Cert.crt

Finish

Cancel

Repeat the process for the Root CA certificate but in the Trusted Root Certificates Folder instead of the Personal Folder. Confirm everything is correct by right clicking the server certificate and checking the certification path.



With this the Certificate creation and installation part of configuring a VPN to work is complete.

Configuring Portal and Gateway

Navigate to Device > Authentication Profile and create an Authentication Profile and make sure the authentication database is local.

Authentication Profile ?

Name: Local_Auth

Authentication | Factors | Advanced

Type: Local Database

User Domain: |

Username Modifier: %USERINPUT%

Single Sign On

Kerberos Realm: |

Kerberos Keytab: Click "Import" to configure this field [X Import](#)

OK Cancel

Switch to the Advanced tab and allow all on the allow list.

Authentication Profile

NameLocal_Auth

AuthenticationFactorsAdvanced

Allow List

☐ ALLOW LIST ^

☒ all

+ Add

- Delete

Account Lockout

Failed Attempts[0 - 10]

Lockout Time (min)0

OK

Cancel

Next, navigate to Network > Interfaces > Tunnel and create a tunnel interface, all a number after the name you configure the tunnel so it can function properly and assign it a separate zone for VPN traffic. This will prompt you to create a new zone.

Tunnel Interface

Interface Nametunnel.10

Comment

Netflow ProfileNone

ConfigIPv4IPv6Advanced

Assign Interface To

Virtual Routerdefault-vr

Security ZoneVPN

OK

Cancel

Zone

?

Name

VPN

Log Setting

None

Type

Layer3

INTERFACES

+

Add

-

Delete

Zone Protection

Zone Protection Profile

None

☒ Enable Packet Buffer Protection

User Identification ACL

☐ Enable User Identification

INCLUDE LIST

^

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

+

Add

-

Delete

Users from these addresses/subnets will be identified.

EXCLUDE LIST

^

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

+

Add

-

Delete

Users from these addresses/subnets will not be identified.

Device-ID ACL

☐ Enable Device Identification

INCLUDE LIST

^

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

+

Add

-

Delete

Devices from these addresses/subnets will be identified.

EXCLUDE LIST

^

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

+

Add

-

Delete

Devices from these addresses/subnets will not be identified.

OK

Cancel

Next go to Network > GlobalProtect > Portals and hit add and make sure to add an interface that is reachable from anywhere globally, most likely an IPv4 Globally Routable Address, if the interface has a static IP address add it in the IPv4 address field.

GlobalProtect Portal Configuration

?

General

Authentication

Portal Data Collection

Agent

Clientless VPN

Satellite

Name

VPN_Portal

Network Settings

Interface

ethernet1/1

IP Address Type

IPv4 Only

IPv4 Address

None

Appearance

Portal Login Page

factory-default

Portal Landing Page

factory-default

App Help Page

None

Log Settings

☐ Log Successful SSL Handshake

☒ Log Unsuccessful SSL Handshake

Log Forwarding

None

OK

Cancel

Next, navigate to authentication and select the TLS/SSL service profile created before.

GlobalProtect Portal Configuration ?

General

Authentication

Portal Data Collection

Agent

Clientless VPN

Satellite

Server Authentication

SSL/TLS Service Profile VPN_SSL/TLS_Profile

Client Authentication

☐	NAME	OS	AUTHENTIC...	AUTO RETRIEVE PASSCODE	USERNAME LABEL	PASSWORD LABEL	AUTHENTI...	ALLOW AUTHENTI... WITH USER CREDENTI... OR CLIENT CERTIFICA...
--------------------------	------	----	--------------	------------------------------	-------------------	-------------------	-------------	---

+ Add - Delete + Clone ↑ Move Up ↓ Move Down

Certificate Profile None

OK Cancel

Next, add an authentication profile. Click add select the authentication profile created previously and make sure to only require the username and password.

Client Authentication ?

Name LocalAuth

OS Any

Authentication Profile vpn authentication profile

☐ Automatically retrieve passcode from SoftToken application

GlobalProtect App Login Screen

Username Label Username

Password Label Password

Authentication Message Enter login credentials

Authentication message can be up to 256 characters.

Allow Authentication with User Credentials OR Client Certificate Yes (User Credentials OR Client Certificate Required)

To enforce client certificate authentication, you must also select the certificate profile in the Client Authentication configuration.

OK Cancel

After everything is configured, it should look like this:

GlobalProtect Portal Configuration ?

General
Authentication
Portal Data Collection
Agent
Clientless VPN
Satellite

Server Authentication

SSL/TLS Service Profile VPN_SSL/TLS_Profile

Client Authentication

	NAME	OS	AUTHENTIC... PROFILE	AUTO RETRIEVE PASSCODE	USERNAME LABEL	PASSWORD LABEL	AUTHENTI... MESSAGE	ALLOW AUTHENTI... WITH USER CREDENTI... OR CLIENT CERTIFICA...
☒	vpn client authentic profile	Any	vpn authentication profile	☐	Username	Password	Enter login credentials	Yes

+ Add
- Delete
Clone
Move Up
Move Down

Certificate Profile None

OK Cancel

Now Navigate to the “Agent” tab, select the root cert to encrypt and decrypt.

Configs ?

Authentication
Config Selection Criteria
Internal
External
App
HIP Data Collection

Name Client Config

Client Certificate None

The selected client certificate including its private key will be installed on client machines.

Save User Credentials Yes

Authentication Override

☐ Generate cookie for authentication override
☐ Accept cookie for authentication override

Cookie Lifetime Hours 24

Certificate to Encrypt/Decrypt Cookie Root-Cert

Components that Require Dynamic Passwords (Two-Factor Authentication)

☐ Portal
☐ External gateways-manual only
☐ Internal gateways-all
☐ External gateways-auto discovery

Select the options that will use dynamic passwords like one-time password (OTP) to authenticate users as opposed to using saved credentials. As a result, the user will always be prompted to enter new credentials for each selected option.

OK Cancel

Now navigate to the external tab, add an external gateway, and put the IP address of the outside facing interface in the address field.

External Gateway?

Name

VPN Address

Address

☐ FQDN
☒ IP

IPv4

192.168.40.167

IPv6

1 item

→

×

☐	SOURCE REGION	PRIORITY
☒	Any	Highest

+

 Add

−

 Delete

☐ Manual (The user can manually select this gateway)

OK

Cancel

Now navigate to the app tab and ensure User Logon is always on.

Configs?

Authentication
Config Selection Criteria
Internal
External
App
HIP Data Collection

App Configurations

Connect Method	User-logon (Always On)
GlobalProtect App Config Refresh Interval (hours)	24 [1 - 168]
Allow user to disconnect GlobalProtect App (Always-on mode)	Allow
Display the following reasons to disconnect GlobalProtect (Always-on mode)	
Allow User to Uninstall GlobalProtect App (Windows Only)	Allow
Allow User to Upgrade GlobalProtect App	Allow with Prompt
Allow user to Sign Out from GlobalProtect App	Yes
Allow user to extend GlobalProtect User Session	No

Welcome Page

None

Disconnect GlobalProtect App (Always-on mode)

Passcode

Confirm Passcode

Max Times User Can Disconnect

0

Disconnect Timeout (min)

0

Uninstall GlobalProtect App

Uninstall Password

Confirm Uninstall Password

Mobile Security Manager Settings

Mobile Security Manager

Enrollment Port

443

OK

Cancel

Click OK to confirm the changes and navigate to the Agent Tab. Under Trusted Root CA select the root certificate and check the Install in Local Root Certificate Store box.

GlobalProtect Portal Configuration ?

General
Authentication
Portal Data Collection
Agent
Clientless VPN
Satellite

Agent

☐	CONFIGS	USER/USER GROUP	OS	EXTERNAL GATEWAYS	CLIENT CERTIFICATE

+ Add
- Delete
Clone
↑ Move Up
↓ Move Down

☐	TRUSTED ROOT CA	INSTALL IN LOCAL ROOT CERTIFICATE STORE
☒	Root-Cert	☐

+ Add
- Delete

Agent User Override Key
Confirm Agent User Override Key

OK Cancel

Click OK to save the GlobalProtect portal configuration.

Configuring the Gateway

Navigate to Network > GlobalProtect > Gateways > Add

Under the general tab, name the gateway and put the IP address of the outside interface.

GlobalProtect Gateway Configuration ?

General
Authentication
Agent
Satellite

General

Name

Network Settings

Interface
IP Address Type
IPv4 Address

Log Settings

☐ Log Successful SSL Handshake
☒ Log Unsuccessful SSL Handshake
Log Forwarding

OK Cancel

Navigate to the authentication page and add client authentication using the authentication profile previously made and make sure the user can login with only username and password.

Client Authentication?

Name

VPN_ClientAuth

OS

Any

Authentication Profile

vpn authentication profile

☐

Automatically retrieve passcode from SoftToken application

GlobalProtect App Login Screen

Username Label

Username

Password Label

Password

Authentication Message

Enter login credentials

Authentication message can be up to 256 characters.

Allow Authentication with User Credentials OR Client Certificate

Yes (User Credentials OR Client Certificate Required)

To enforce client certificate authentication, you must also select the certificate profile in the Client Authentication configuration.

OK

Cancel

Navigate to Agent > Tunnel Settings, select the tunnel interface and make sure to select the enabled IPsec option.

GlobalProtect Gateway Configuration?

General

Authentication

Agent

Satellite

Tunnel Settings

Client Settings

Client IP Pool

Network Services

Connection Settings

Video Traffic

HIP Notific

☒ Tunnel Mode

Tunnel Interface

tunnel

Max User

[1 - 250]

☒ Enable IPsec

GlobalProtect IPsec Crypto

default

☐ Enable X-Auth Support

Group Name

Group Password

Confirm Group Password

☒ Skip Auth on IKE Rekey

OK

Cancel

Go to client settings > add, make the config selection criteria for all fields.

Configs

Config Selection Criteria

Authentication Override

IP Pools

Split Tunnel

Network Services

Name

VPN Selection Criteria

Config Selection Criteria

any

Any

SOURCE USER

OS

Add

Delete

Add

Delete

Source Address

REGION

IP ADDRESS

Add

Delete

Add

Delete

The configuration must match User and OS and either Region or IP Address if specified.

OK

Cancel

Now go to IP pools and add an IP pool separate from any other IP pools used for internal users; this IP will be the tunnel IP address.

Configs

Config Selection Criteria

Authentication Override

IP Pools

Split Tunnel

Network Services

Retrieve Framed-IP-Address attribute from authentication server

AUTHENTICATION SERVER IP POOL

Enter IP subnets or ranges to match the Framed IP attribute of the authentication server. Supports IPv4 private/public addresses (e.g. 192.168.74.0/24, 192.168.75.1-192.168.75.100) or IPv6 unique local/public addresses (e.g. 2001:aa::1-2001:aa::10)

Add

Delete

These IPs will be added to the firewall's routing table

IP POOL

192.168.2.0/24

Add

Delete

Move Up

Move Down

These IPs will be added to the firewall's routing table

OK

Cancel

Finally, navigate to split tunnel and make sure nothing is there unless you want to limit what VPN users can reach from the internet.

Configs

Config Selection Criteria

Authentication Override

IP Pools

Split Tunnel

Network Services

Access Route

Domain and Application

☐ No direct access to local network
 No direct access to local network is applicable to Windows, Mac and Linux only

☐ INCLUDE ^ Enter subnets that clients need to access (e.g. 172.16.1.0/24 or 21DA:D3:0:2f3B::/64)	☐ EXCLUDE ^ Enter subnets that clients should exclude (e.g. 172.16.1.0/24)
+ Add - Delete	+ Add - Delete

These routes will be added to the client's routing table. More-specific routes take precedence over less-specific routes.

OK

Cancel

Add a local user to Device > Local User > Database > Users, this name will be the username to log on to the firewall and so will the password.

Local User

Name

User

Mode

☒ Password
 ☐ Password Hash

Password

.....

Confirm Password

.....

☒ Enable

OK

Cancel

Now, commit all changes. Once committed on another device connect to the VPN by typing the outside interface IP address and download the software client after logon. Then open the software, enter the portal address, and connect to the VPN.

Problems

I first encountered a problem when configuring the certificates as I initially only installed the root certificate into the client machine and forgot to select the option when configuring the portal and gateway to require either the login credentials or have the necessary certificates which made it impossible to login to the portal.

To fix this, I installed the server level certificate in the personal folder and checked the hierarchy of the server and root certificate to mirror the hierarchy of the certificates in the firewall. Then, I configured the firewall to accept login credentials by themselves.

I then tried logging into the firewall but did not create a local user, so when I tried to log in using the credentials to log into the firewall, I was not allowed in. This is because of the authentication profile used in the portal and gateway configuration was local authentication as we did not set up any authentication servers like RADIUS. To fix this, I configured local users in the firewall configuration so that anyone with the same login credentials as the created user would be able to log into the portal.

After these fixes, I was able to successfully log into the portal, download the client, and connect to the gateway and successfully access resources from inside the network.

Conclusion

This lab was great in helping me gain a strong foundational understanding of how VPNs operate and how to configure them using a Palo Alto 220, and GlobalProtect. This lab has also helped gain more proficiency in configuring Palo Alto Devices and VPN configurations in general, adding more skills within my existing skillsets. I hope to further develop my skills in VPN deployment and configuration through other firewalls and potentially create my own VPN using a docker.